MIL-I-60635(MU)
31 May 1968
SUPERSEDING
FA-PD-MI-2244, Rev. 0
15 November 1960

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, M9 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, M9, Assembly, loaded with Cartridge, Initiator, M69 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

F10523303	- Initiator, Cartridge Actuated, M9, Assembly
C8797660	- Box, Packing, Ammunition for Initiator, Cartridge Actuated, M9 in Fiber Container, M381

FSC 1377

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2.1 DRAWINGS (Cont'd)

Munitions Command
C8797661 - Ammunition Container, Fiber, M381 for Initiator, Cartridge Actuated, M9
IEL-10523303 - Inspection Equipment List for Initiator, Cartridge Actuated, M9, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with all the requirements specified on the applicable drawings and specifications.

3.2 Breakaway torque.- After the initiator has been completely assembled, the threaded joint between the head and body and between the cap and body shall withstand a breakaway torque of 400 inch-pounds.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3° in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 Ballistics.- Initiators loaded with accepted cartridges shall produce pressure values specified in Table I. Pressure values shall be produced at the end of 30 foot length of aircraft hose, size No. 4, as specified in MS28741.

TABLE I

Temperature	Pressure (PSI) Min.	Max.	Ignition Delay Max. (See 6.3)
-65° to -70°F	1000	2200	65 milliseconds
65° to 75°F	1200	2300	65 milliseconds
160° to 165°F	1300	2400	65 milliseconds

3.4.2 Initiation.- The initiator shall be fired by a source delivering an output equal to that of an M3A2 initiator at the end of 20 feet of aircraft hose, size No. 4, as specified in MS28741, at 70° ± 5F.

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3.4.3 The initiator shall not fail to fire.

3.4.4 The initiator shall function without mechanical failure (See 6.2).

3.5 Workmanship.- Workmanship requirements are as specified on the applicable drawings, referenced specifications and the following.

3.5.1 Metal defects.- The metal parts shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

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4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number in Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, M69, Drawing D8593496, covering a detail of Drawing F10523303

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

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4.2.2.2 Assembly, Drawing F10523303, less Drawings D10522666, F10522493, C10522667, C10522670, C10522671 and D8593496

Defect	Method of Inspection
CRITICAL	
1. Firing pin missing	Visual
2. Firing pin incorrectly assembled	Visual
3. Shear pin missing	Visual
4. "O" ring missing	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. "O" ring not lubricated	Visual

4.2.2.3 Assembly, Drawing F10523303, less Drawings D10522668, C10522494, B10522495, C10522670 and C10522671

CRITICAL	
1. Filter missing or improperly assembled	Visual
2. Cartridge missing	Visual
3. Cartridge loose	Visual and Manual
4. "O" ring missing	Visual
MAJOR	
101. Any part missing, except as otherwise classified	Visual
102. Thread damaged	Visual
103. Foreign matter	Visual
MINOR	
201. "O" ring not lubricated	Visual

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4.2.2.4 Assembly, Drawing F10523303 less Drawings C10522670 and C10522671

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	Visual
2. Any part missing, except as otherwise classified	Visual
3. Marking missing, incorrect or illegible	Visual
4. Foreign matter in gas port	Visual
MAJOR	
101. Thread damaged	Visual
MINOR	
201. Location of marking incorrect	Visual
202. Sealing compound missing from set screws	Visual
203. Copalite missing from body threads	Visual
204. Shipping plus or cap missing	Visual
205. Protective finish inadequate or defective	Visual
206. Foreign matter	Visual

4.2.2.5 Assembly, Drawing F10523303

CRITICAL	
None defined	
MAJOR	
101. Assembly of block and strap insecure or incorrect	Visual and Manual
MINOR	
None defined	

4.2.2.6 Packing in Fiber Container, Drawing F8797661

CRITICAL	
None defined	

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing in Wooden Box, Drawing F8797660

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual and Manual
203. Strapping missing, loose, broken or improperly assembled	Visual and Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- The breakaway torque shall be applied to each initiator at the threaded joint of the body-head and body-cap to determine compliance with 3.2. Failure of any initiator to comply with the requirements of 3.2 shall be cause for rejection of the initiator.

4.2.3.2 Residual magnetism.- Each initiator shall be tested for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 Functional.- The following test shall be performed by the Government.

4.2.3.3.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.4.1, 3.4.3 and 3.4.4. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements of 3.4.1, 3.4.3 and 3.4.4 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
160° to 165°F	7

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10523303.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.- A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with a fixture to determine the adequacy of the joints. The body shall be secured in an air vise and the torque shall be applied to the head and to the cap.

4.3.2 Residual magnetism.- The initator shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

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4.3.3 Functional.

4.3.3.1 Ballistics.- The sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber.

4.3.3.1.1 Temperature conditioning.- A standard M9 Initiator, loaded with a dummy M69 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature. Sample shall be reconditioned if the 5 minute time for firing or [illegible] is exceeded.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings F8797660 and F8797661.

5.2 Data card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 "Ignition delay" is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

6.4 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2244, Rev. 0 dated 15 November 1960.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A153

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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